

★ Note to Professor: I use ⚡ to show where the comparison fails, similar to the Blackboard notes and the textbook.

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Problem Set 5

32.1-1 Show the comparisons the naive string matcher makes for the pattern $P=0001$ in the text:

$T=000010001010001$

$S=0: 000010001010001$

$\begin{array}{c} \text{0001} \\ \text{0001} \end{array}$ Not a match

$S=1: 000010001010001$

$\begin{array}{c} \text{0001} \\ \text{0001} \end{array}$ This is a match

$S=2: 000010001010001$

$\begin{array}{c} \text{0001} \\ \text{0001} \end{array}$ Not a match

$S=3: 000010001010001$

$\begin{array}{c} \text{0001} \\ \text{0001} \end{array}$ Not a match

$S=4: 000010001010001$

$\begin{array}{c} \text{0001} \\ \text{0001} \end{array}$ Not a match

$S=5: 000010001010001$

$\begin{array}{c} \text{0001} \\ \text{0001} \end{array}$ This is a match

$S=6: 000010001010001$

$\begin{array}{c} \text{0001} \\ \text{0001} \end{array}$ Not a match

$S=7: 000010001010001$

$\begin{array}{c} \text{0001} \\ \text{0001} \end{array}$ Not a match

$S=8: 000010001010001$

$\begin{array}{c} \text{0001} \\ \text{0001} \end{array}$ Not a match

32.1-1 (cont.)

$s=9$: 00001000100001

0001

Not a match

$s=10$: 00001000100001

0001

Not a match

$s=11$: 00001000100001

0001

This is a match

There are 3 valid shifts for the pattern P in the text T . These valid shifts occur when $s=1$, $s=5$, and $s=11$.

★ Note to Professor: This symbol ";" is my way of writing a semicolon. It is not the letter lowercase "i". Just thought I'd clarify

(32.2-1) Working modulo $q=11$, how many spurious hits does the Rabin-Karp matcher encounter in the text $T=3141592653589793$ when looking for the pattern $P=26$?

First, let's do: $26 \bmod 11 = 4$. Now, let's go through T :

$s=0$: 3141592653589793

↓ mod 11

t_0 : 9 ; $9 \neq 4$, so not valid

$s=1$: 3141592653589793

↓ mod 11

t_1 : 3 ; $3 \neq 4$, so not valid

$s=2$: 3141592653589793

↓ mod 11

t_2 : 8 ; $8 \neq 4$, so not valid

$s=3$: 3141592653589793

↓ mod 11

t_3 : 4 ; $4 = 4$, but $15 \neq 26$, so this is a spurious hit

$s=4$: 3141592653589793

↓ mod 11

t_4 : 4 ; $4 = 4$, but $59 \neq 26$, so this is a spurious hit

$s=5$: 3141592653589793

↓ mod 11

t_5 : 4 ; $4 = 4$, but $92 \neq 26$, so this is a spurious hit

$s=6$: 3141592653589793

↓ mod 11

t_6 : 4 ; $4 = 4$, and $26 = 26$, so this is a valid hit

$s=7$: 3141592653589793

↓ mod 11

t_7 : 10 ; $10 \neq 4$, so not valid

$s=8$: 3141592653589793

↓ mod 11

t_8 : 9 ; $9 \neq 4$, so not valid

32.2-1 (cont.)

$S=9$: 3 1 4 1 5 9 2 6 5 3 5 8 9 7 9 3

↓ mod 11

t_9 : 2; $2 \neq 4$, so not valid

$S=10$: 3 1 4 1 5 9 2 6 5 3 5 8 9 7 9 3

↓ mod 11

t_{10} : 3; $3 \neq 4$, so not valid

$S=11$: 3 1 4 1 5 9 2 6 5 3 5 8 9 7 9 3

↓ mod 11

t_{11} : 1; $1 \neq 4$, so not valid

$S=12$: 3 1 4 1 5 9 2 6 5 3 5 8 9 7 9 3

↓ mod 11

t_{12} : 9; $9 \neq 4$, so not valid

$S=13$: 3 1 4 1 5 9 2 6 5 3 5 8 9 7 9 3

↓ mod 11

t_{13} : 2; $2 \neq 4$, so not valid

$S=14$: 3 1 4 1 5 9 2 6 5 3 5 8 9 7 9 3

↓ mod 11

t_{14} : 5; $5 \neq 4$, so not valid

There were a total of 3 spurious hits when working modulo $q=11$ for the text T and pattern P .

These spurious hits occur when $S=3$, $S=4$, and $S=5$.